

NAT Network Create

Create clone of VM

Link to NAT Network for both VMs

Update Mac Address of clone

(if neccess) set new DHCP lease

chmod +x setup_client.sh

DEMO #1

OG (client)

② sudo ./vpncclient (T1)

④ bash setup_client.sh (T2)

⑤ ifconfig tun0 (shows tunnel's there) (T2)

⑥ ping 10.4.2.5 (runs) (T3)

⑧ (ping freezes) (T3)

⑩ sudo pkill vpncclient (T2)
step 2 again (T1)

⑫ step 4 again (T2)

⑬ step 6 again (T3)

clone (server)

① sudo ./vpnserv (T1)

③ bash setup_server.sh (T2)

⑦ sudo pkill vpnserv (T2)

⑨ step 1 again (T1)

⑪ step 3 again (T2)

T1

T2

T1

T2

T3

Server

Client

- Canvas
- Widgets
- Pages
- Dashboard
- Assignments
- Notes
- Grades
- Courses
- Calendar
- Inbox
- Help
- ...
- Courses
- EN100_C_D
- CS122A
- 67°F Clear

DEMO #2

[run immediately after ^{demo}#1]

OG (client)

clone (server)

① bash add_route.sh (T2)

③ ping -c 3 10.0.8.1 (T3)
count 3

④ bash remove_route.sh (T2)

⑥ ping 10.0.8.1 (T3)

② sudo tcpdump -i tun0 (T2)
(show traffic is entering tunnel)

⑤ step 2 again (T2)
(no traffic)

DEMO #3

OG (Client)

② sudo ./tapclient (T1)

④ bash setup-tap-client.sh (T2)

⑥ bash demo3-client.sh (T2)

⑦ sudo pkill tapclient (T3)

⑧ ping 10.0.32.2 (T3)

Clone (Server)

① sudo ./tapserver (T1)

③ bash setup-tap-server.sh (T2)

⑤ sudo tcpdump -i tap0 -e (T2)

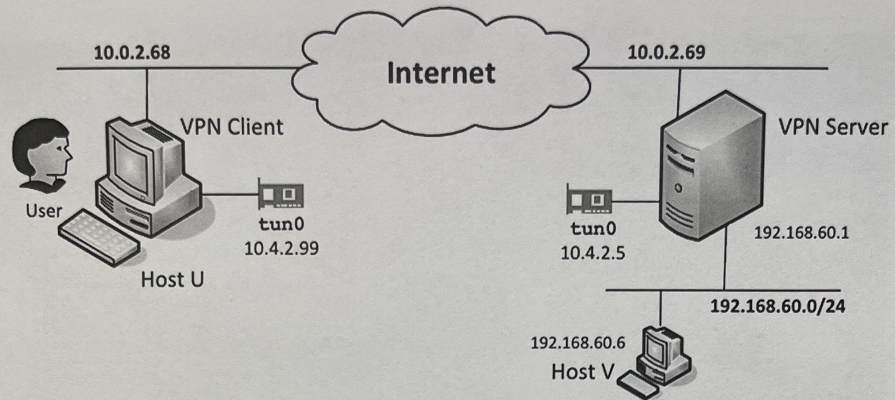


Figure 19.7: Network setup for a VPN.

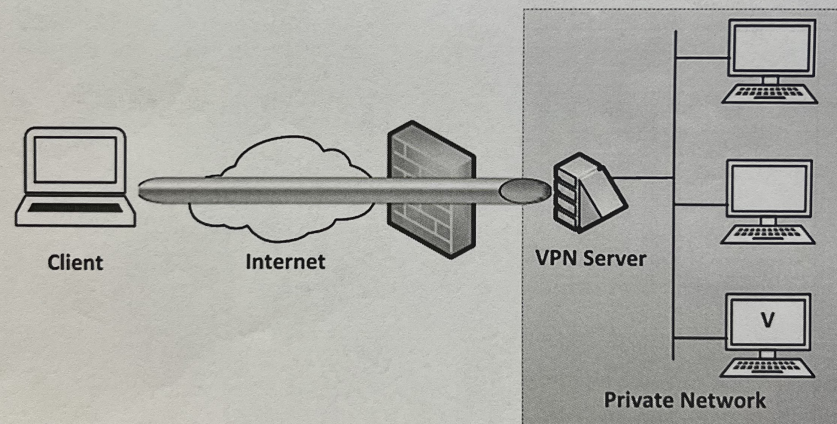
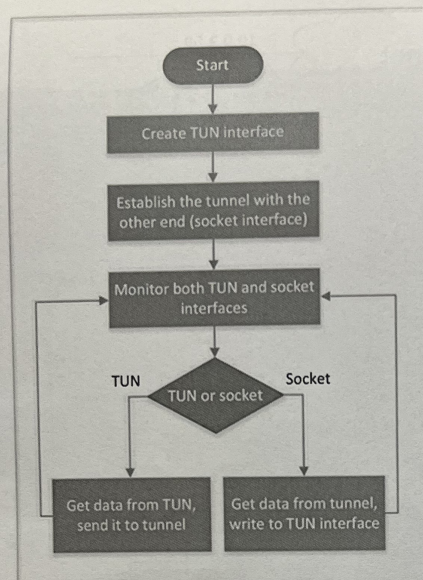
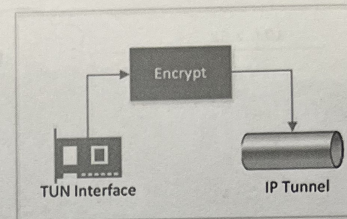


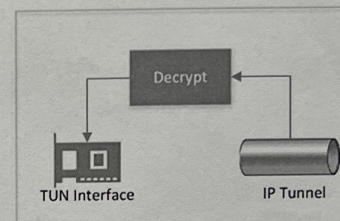
Figure 19.1: Overview of a Virtual Private Network.



(a) The Flow of the Program



(b) From TUN Interface to Tunnel



(c) From Tunnel to TUN Interface

Figure 19.6: How a sample VPN program is implemented